

Unit 2 – Exploring Two Variable Data

Notes 2 – Scatterplots and Correlation

CATEGORICAL

QUANTITATIVE

We have worked with data for bar graphs, box plots, dot plots, and histograms. The type of data that is represented with these graphs are UNIVARIATE data.

In our Vitruvian Man activity, we explored BIVARIATE data. When we compare two variables, we are exploring the relationship between them.

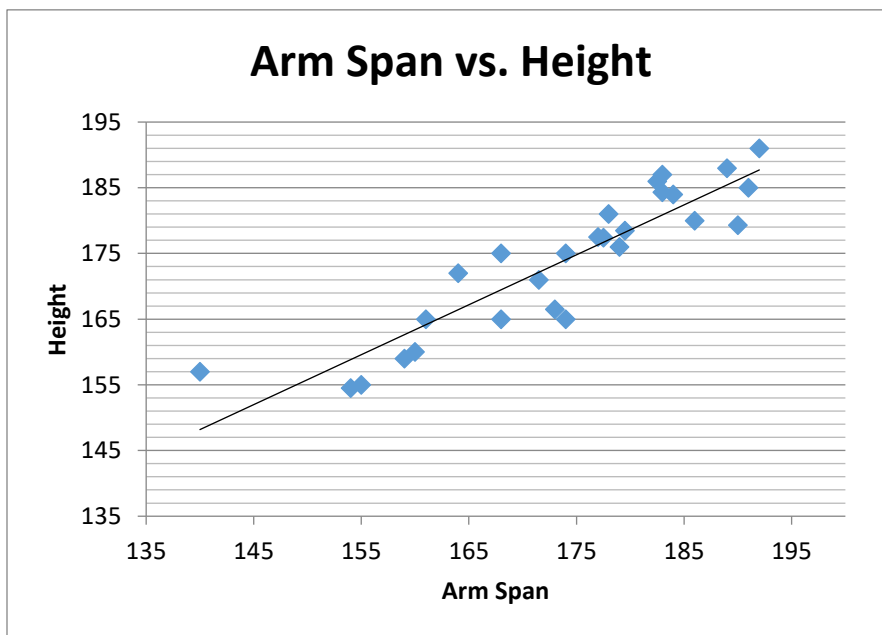
Most statistical studies involve more than one variable. Often in the AP Statistics exam, you will be asked to compare two data sets by using side by side boxplots or histograms etc. However, there are times where we want to examine relationships among several variables for the same group of data (in the Vitruvian Man activity, each person had TWO sets of data: height and arm span).

Graphing Two Variables

When you examine the relationship between two variables you need to start with a SCATTERPLOT.

A scatterplot shows the relationship between two quantitative variables measured on the same individuals. The values of one variable appear on the horizontal axis, and the values of the other variable appear on the vertical axis. Each individual in the data appears as a point in the plot fixed by the values of both variables for that individual.

Here is a previous class's data from this activity:



First, we have to identify and name the correct variables we used in this activity.

A RESPONSE variable measures the outcome of a study or an observation.

An EXPLANATORY variable helps explain or influences change in a response variable.

****You will often find explanatory variables called *independent* variables, and response variable called *dependent* variables. The idea behind this language is that the response variable *depends* on the explanatory variable. Because the**

words independent and dependent have other, unrelated meanings in statistics, we won't use them here.

X variable = INDEPENDENT = EXPLANATORY

Y variable = DEPENDENT = RESPONSE

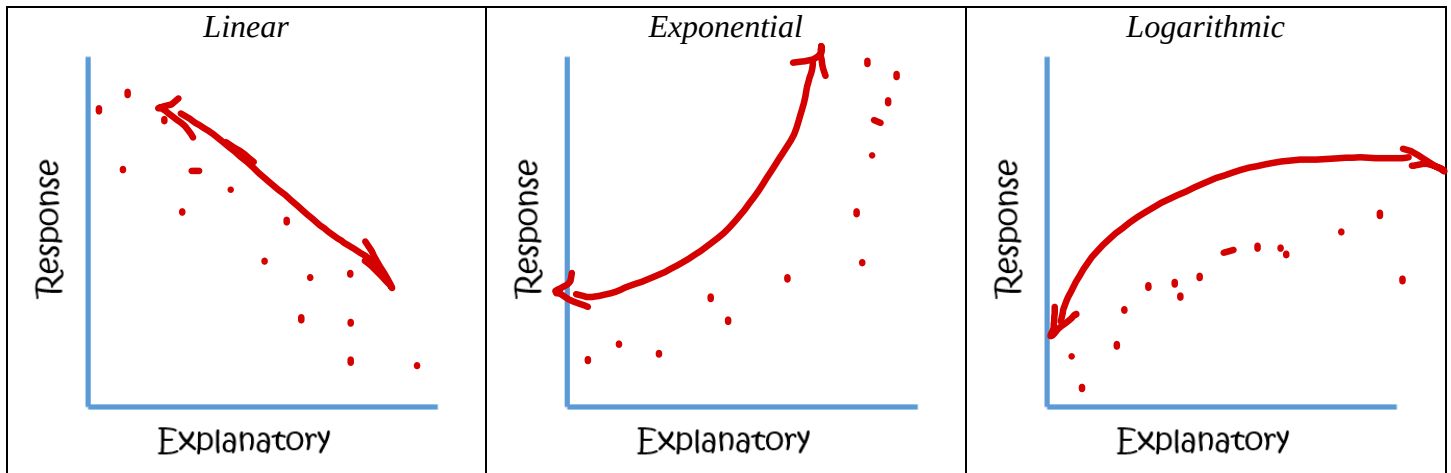
Interpreting a Scatterplot

- In any graph of data, look for overall pattern and for striking deviations from that pattern.
- You can describe the overall pattern of a scatterplot by the FORM, DIRECTION, and STRENGTH of the relationship.
- An important kind of deviation is an OUTLIER, an individual value that falls outside the overall pattern of the relationship.

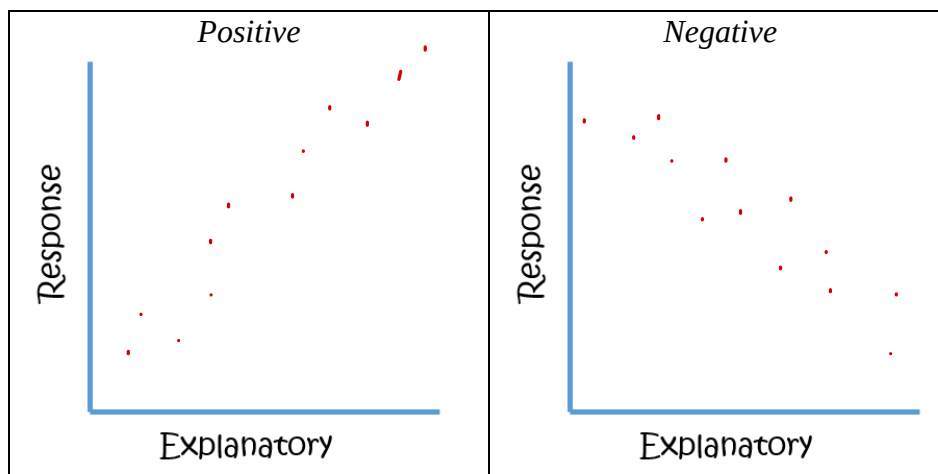
Things to look for in a scatterplot:

- Form: Overall pattern (linear, exponential, etc.) or deviations from the pattern (outliers)
- Direction: Positive or negative slope
- Strength: How close do the points lie to a simple form (such as a line)

Form: THE PATTERN CAN BE RANDOM OR FOLLOW A FUNCTION

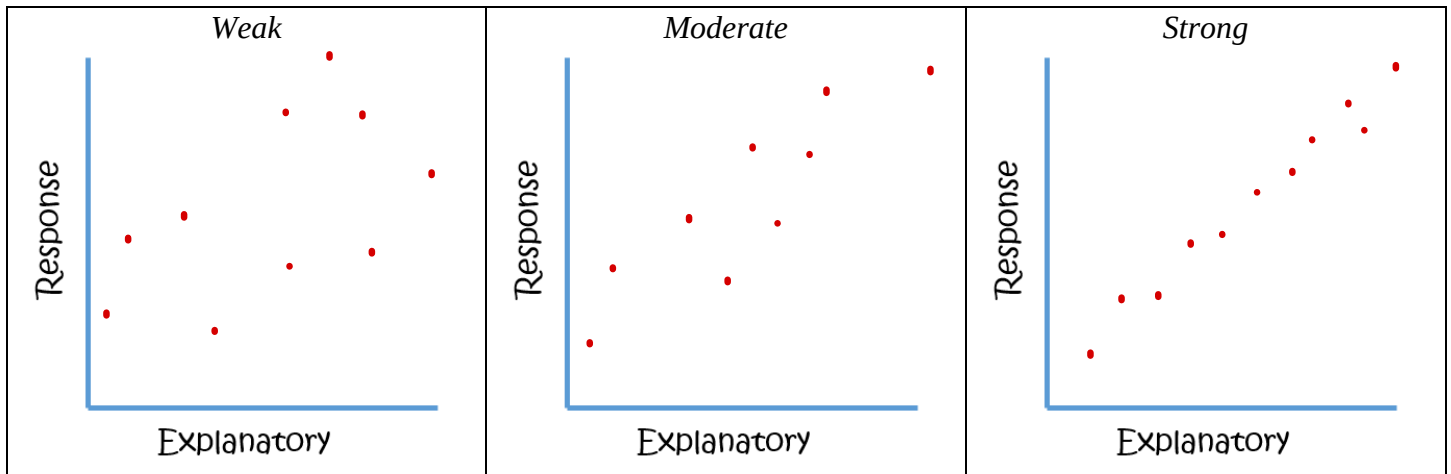


Direction: IS THE FORM TRENDING UP (+) OR DOWN (-) AS YOU MOVE



ALONG THE
X-AXIS.

Strength: *How WELL DO THE POINTS FALL ON A LINE?*



** POSITIVE, LINEAR EXAMPLES*

Example

Suppose we hypothesize that the number of doctor visits a person has can be explained by the amount of cigarettes they smoke. So we want to see if there is a relationship between the number of cigarettes one smokes a week and the number of times per year one visits a doctor. We ask 10 random people and get the following information:

# of Cigarettes Per Week	0	3	21	15	30	5	40	60	0	0
Number of doctor visits per year	1	2	4	3	5	1	5	6	2	0

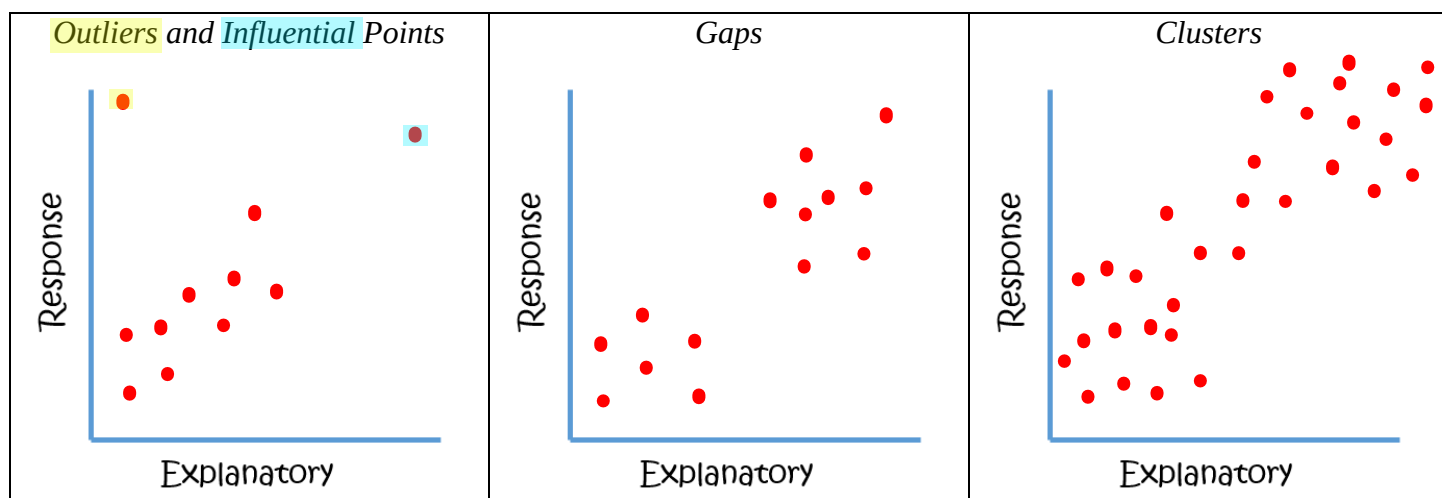
Create a scatterplot and describe the relationship between the variables.

Creating a Scatterplot on the Calculator

Use the example on the previous page to create a scatterplot on the calculator.

- 1) Load the x-values into list 1 and the y-values into list 2.
- 2) Using StatPlot - highlight the mini-scatterplot: XList: L1 and YList: L2
- 3) Press “graph” but you will have to “Zoom 9” to fit the scatterplot on the screen

Unusual Features



Correlation

In order to strengthen the analysis when comparing two variables, we can attach a number, called the correlation coefficient (r), to describe the linear relationship between two variables. This number helps remove any subjectivity in reading a linear scatter plot.

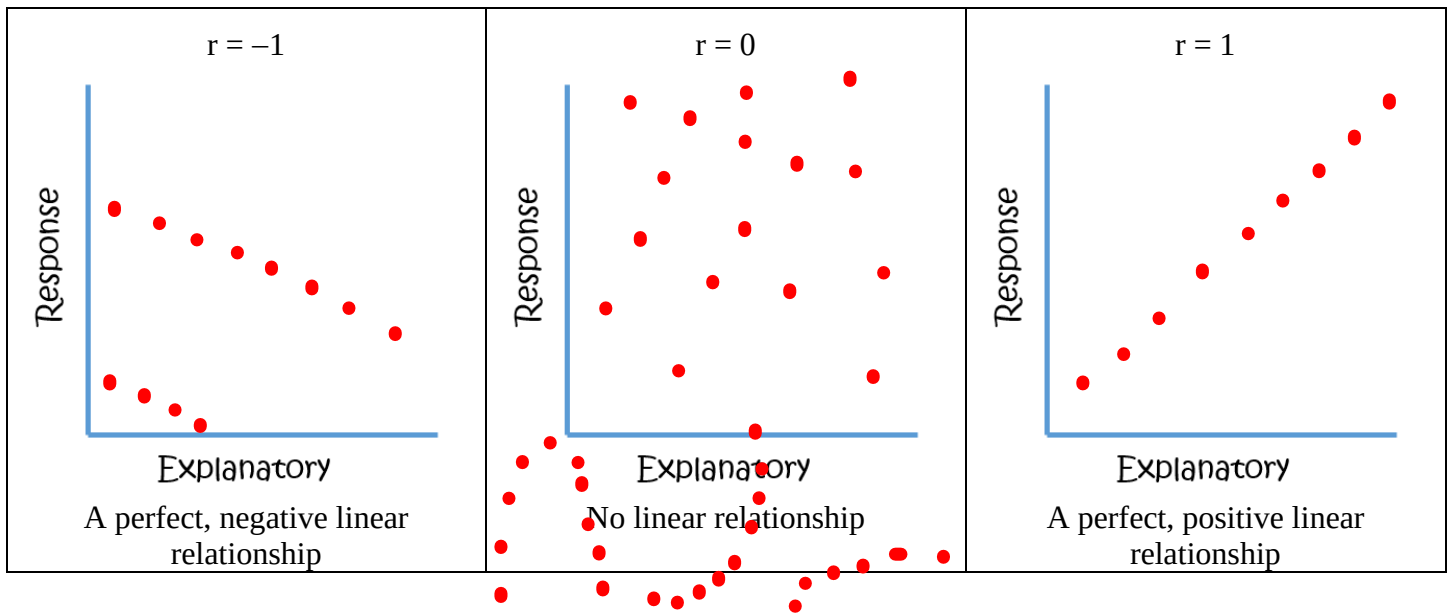
The **r COEFFICIENT** measures the strength and direction of the linear relationship between two quantitative variables.

While we will never have to find correlation by hand, the formula is provided to us on the AP Statistics formula sheet. There are a few facts about the correlation that the formula can help us remember.

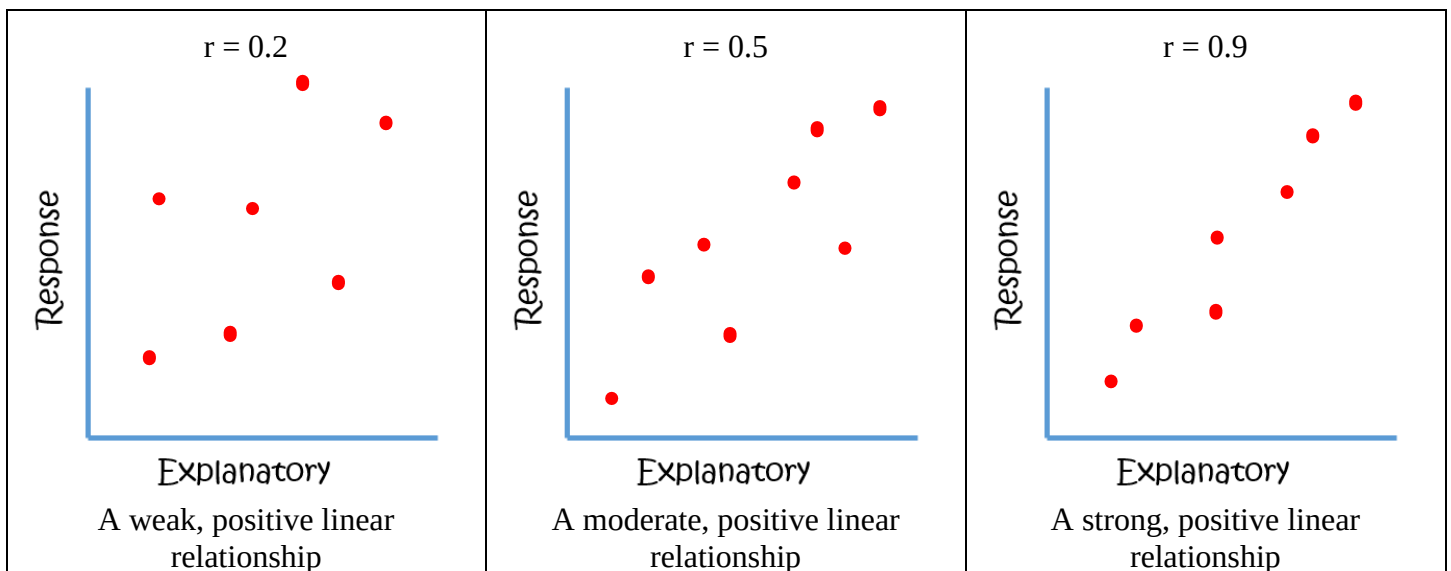
$r = \frac{1}{n-1} \sum \left(\frac{x_i - \bar{x}}{S_x} \right) \left(\frac{y_i - \bar{y}}{S_y} \right)$	r : correlation
	x_i : each x-value
	y_i : each y-value
	$\bar{x}$: mean of the x values
	$\bar{y}$: mean of the y values
	S_x : standard deviation of x values
	S_y : standard deviation of y values

Essentially, the correlation coefficient, r , finds the average of the product of the standardized scores.

Fact #1: Correlation is a number that is between -1 and 1 .

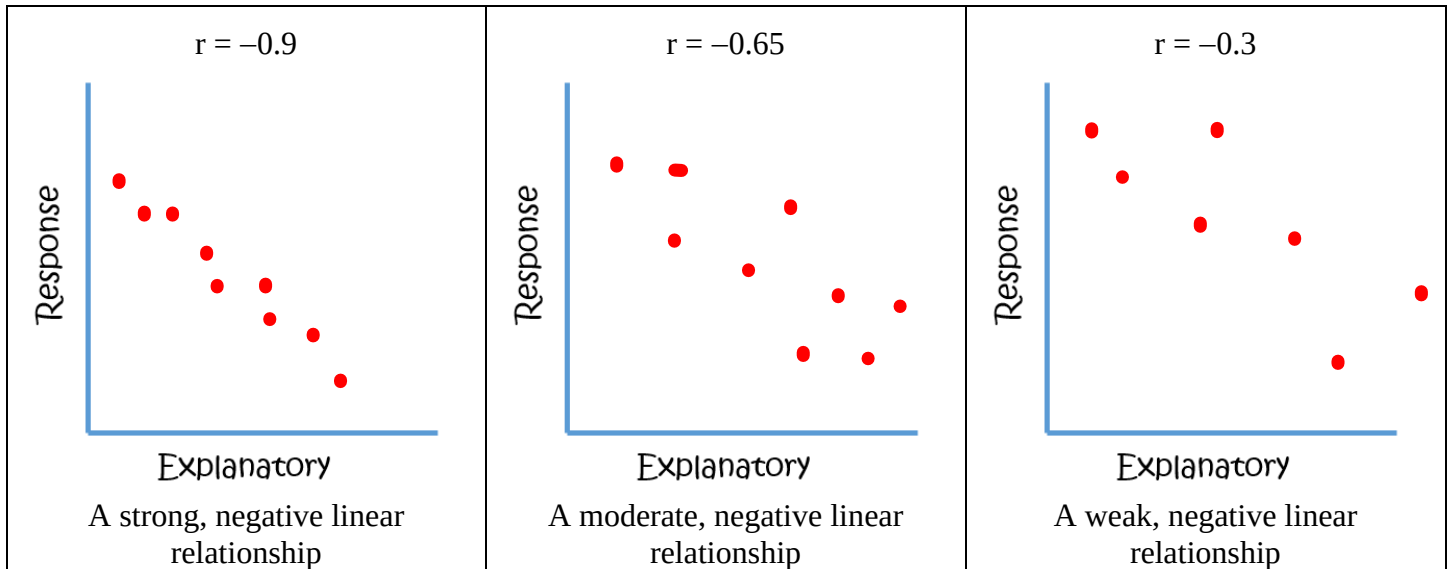


Fact #2: Positive correlations between 0 and 1 have varying strengths, with the strongest positive correlations being closer to 1 .

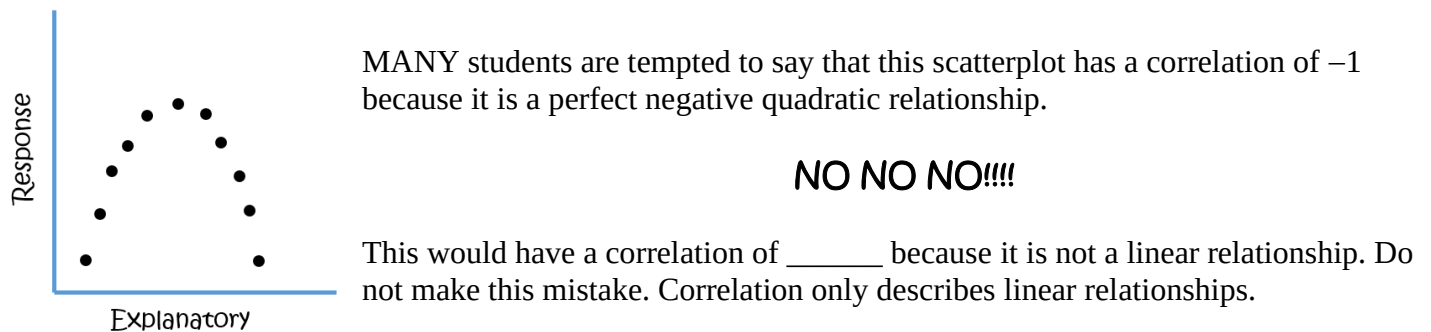


**Note: There is no magic “cutoff number” for describing weak/moderate/strong. Use your statistical intuition!*

Fact #3: Negative correlations between -1 and 0 have varying strengths, with the strongest negative correlations being closer to -1 .



Fact #4: Correlation describes only linear relationships between two variables.



Fact #5: Correlation does not have units and changing units on either axis will not affect correlation.

To see this, take a look at the formula again:

$$r = \frac{1}{n-1} \sum \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$$

Since we are standardizing all the x and y values, it does not matter what the units are! We take the product of their standardized scores. Speaking of that....

Fact #6: Switching the explanatory and response variables on the axes will not change the correlation.

Again, looking at the formula, this is because the order of the multiplication does not matter. Correlation makes no distinction between explanatory and response variables. It makes no difference which variable you call x and which you call y when calculating the correlation.

Fact #7: Correlation is very strongly affected by outliers.

Use correlation with caution when outliers appear in your scatter plot. Don't rely on correlation alone to determine the linear strength between two variables – graph a scatter plot first!